INORGANIC CUMULATIVE EXAM

April 21, 2006

Synthetic Inorganic Chemistry **Do 2 parts for each Question**

1) Several of the top ten industrial chemicals (by quantity and total value) are important basic inorganic compounds. Describe the industrial method for preparing two of the following. Be as specific as possible in your answer, i.e. give the general operating conditions (e.g. high vs low temperature, etc.). Indicate where catalysts are required and if possible the catalyst of choice.

a) H_2SO_4 b) $HN0_3$ c) NH_3 d) H_2 e) your choice but it must be a high volume chemical

2) Frequently laboratory syntheses following different paths than large-scale industrial preparations. Describe a lab scale synthesis for two of the following. Be as specific as possible about reagents and conditions in your discussion.

a) H_2 (g) (different method than in #1) b) HCl (g) c) O_2 (triplet state) d) Br_2

3) Occasionally "exotic" inorganic compounds find use as synthetic reagents. Describe the use of two of the following in this role and indicate the probable basis for their activity.

a) SF_4 b) $Xe0_3$ c) SbF_5/HSO_3F d) $(t\text{-}Bu)_2N^{-1}$ (as a ligand)

4) Account for the frequent use of the following ions in forming salts. What characteristic do these ions show that make them particularly useful in specific instances? I.e., if you had chosen to form a salt using these species as counterions, what reason(s) might have driven the choice? Do two **(a+b choice not accepted)**

a) $C10_4^{-1}$ b) PF_6^{1-} (instead of $CI0_4^{1-}$) c) $(C_2H_5)_4N^{1+}$ d) a counterion of your choice